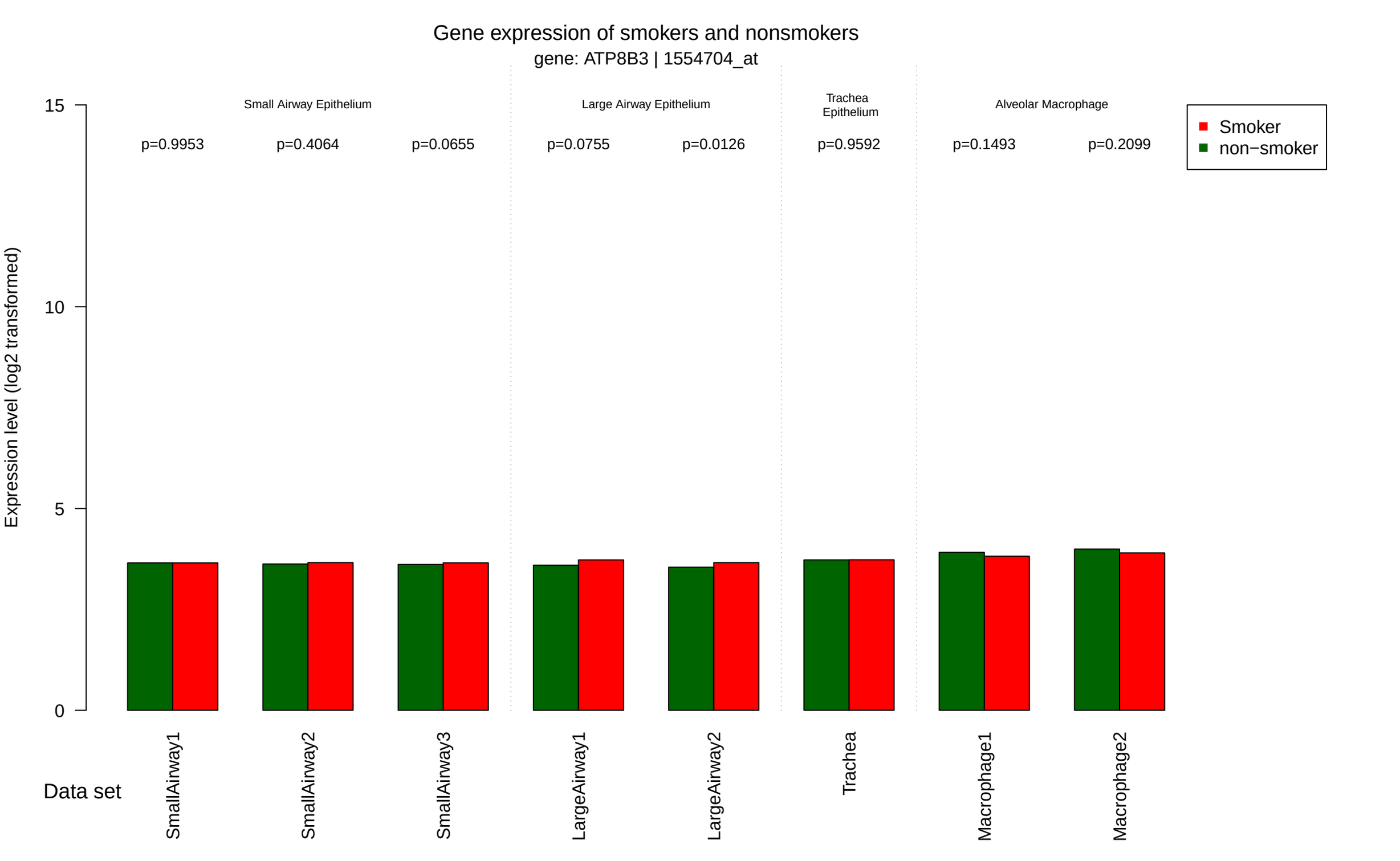
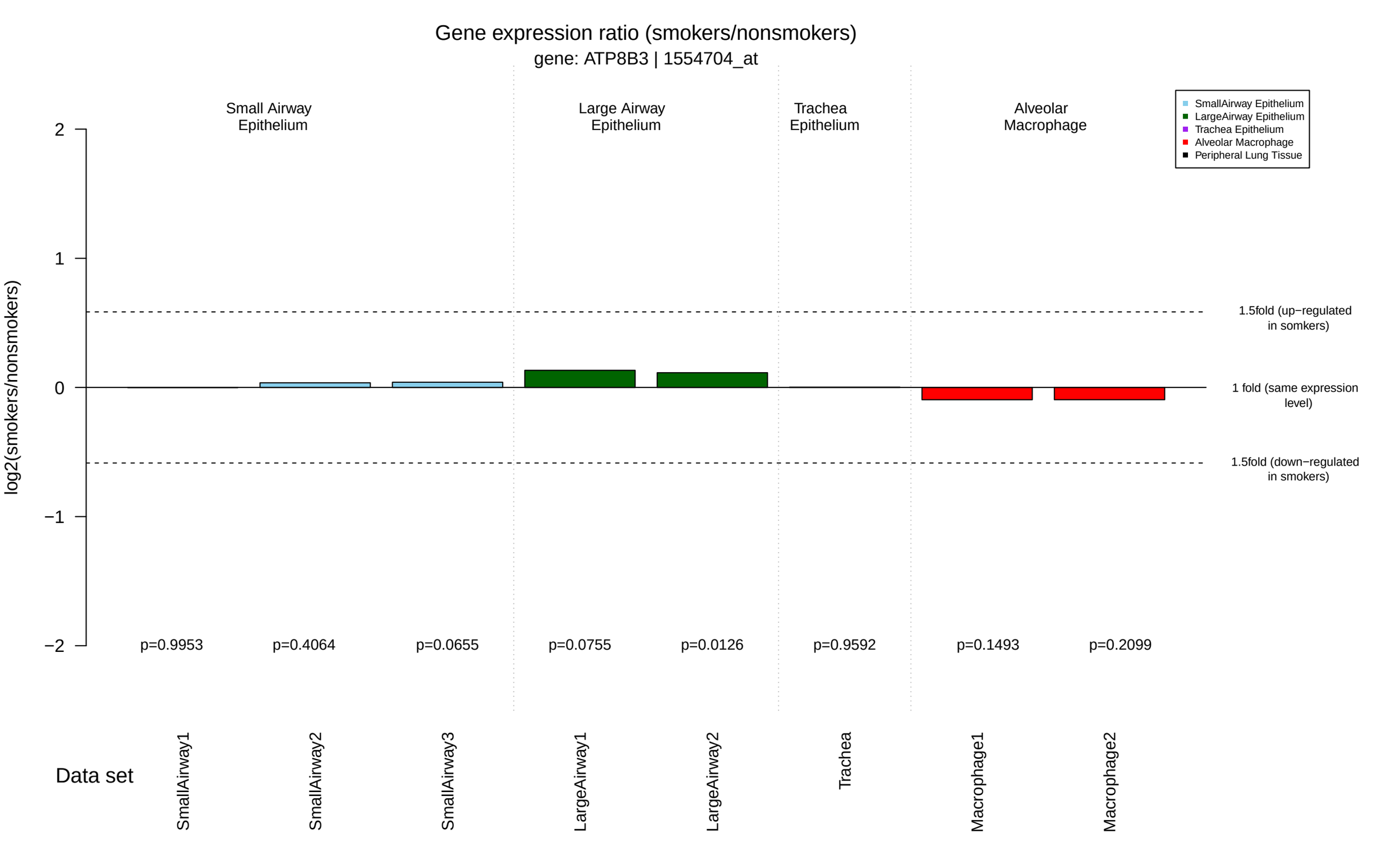
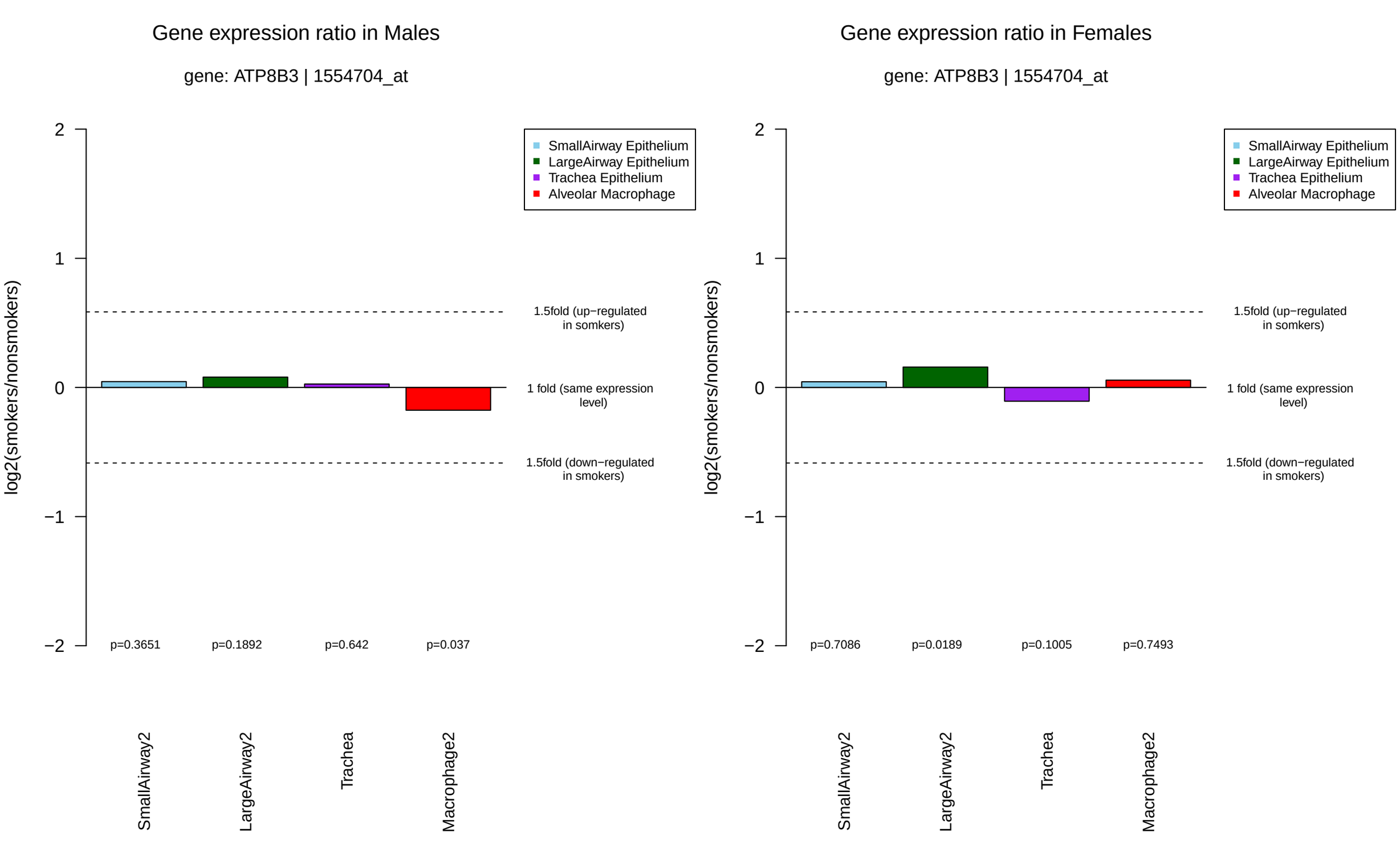
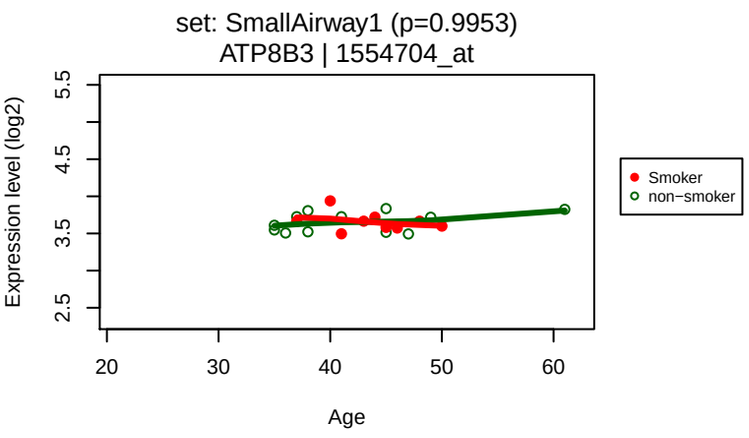
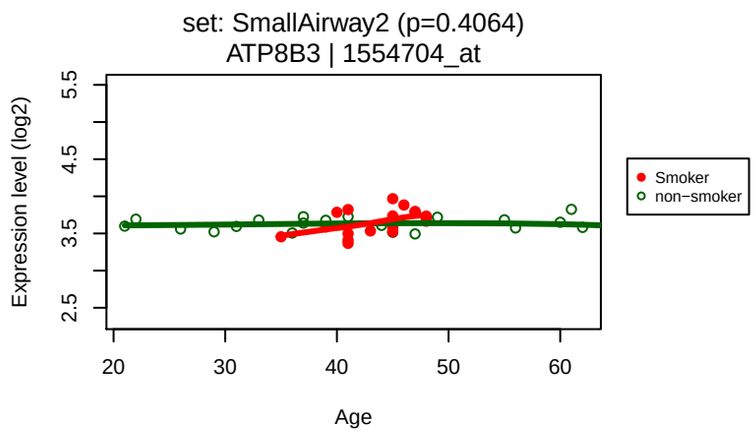
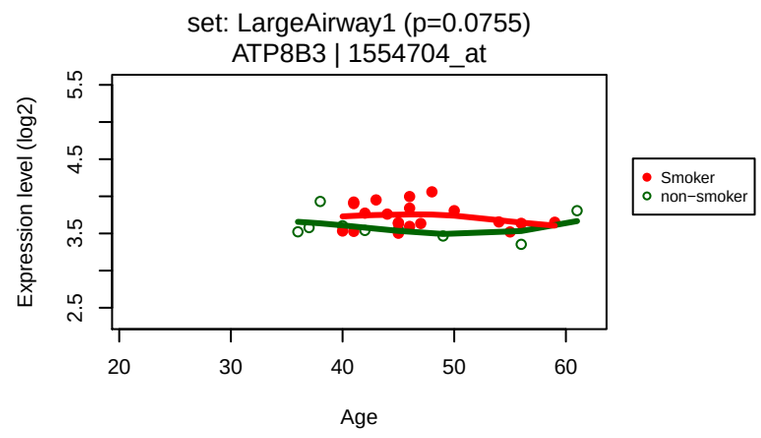

Gene expression of smokers and nonsmokers
gene: ATP8B3 | 1554704_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9953
p=0.4064
p=0.0655
p=0.0755
p=0.0126
p=0.9592
p=0.1493
p=0.2099
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

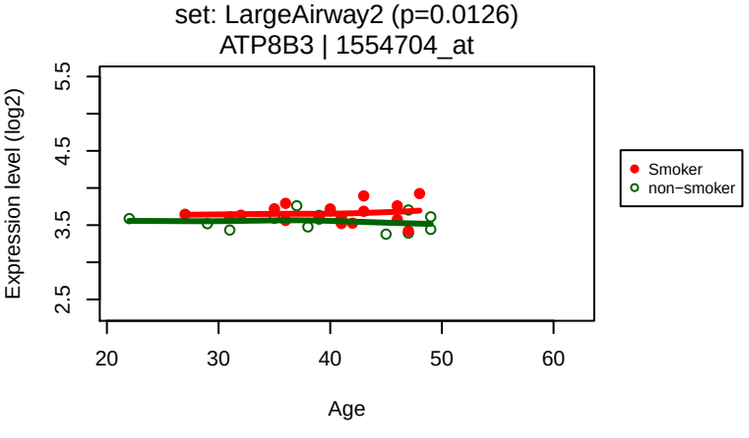

Gene expression ratio (smokers/nonsmokers)
gene: ATP8B3 | 1554704_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9953
p=0.4064
p=0.0655
p=0.0755
p=0.0126
p=0.9592
p=0.1493
p=0.2099
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

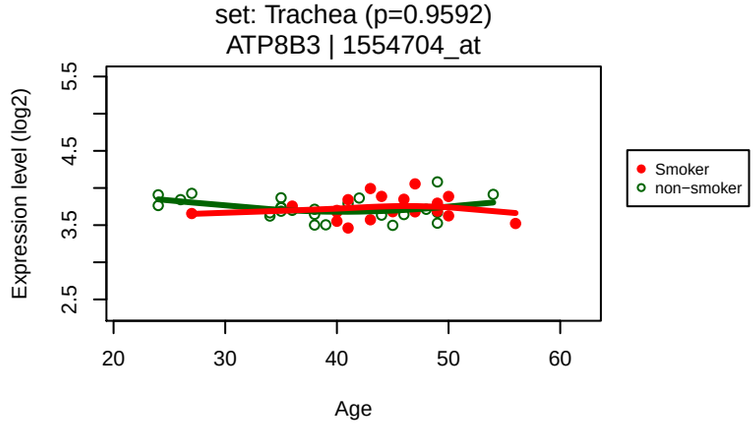

Gene expression ratio in Males
gene: ATP8B3 | 1554704_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3651
p=0.1892
p=0.642
p=0.037
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP8B3 | 1554704_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7086
p=0.0189
p=0.1005
p=0.7493
SmallAirway2
LargeAirway2
Trachea
Macrophage2

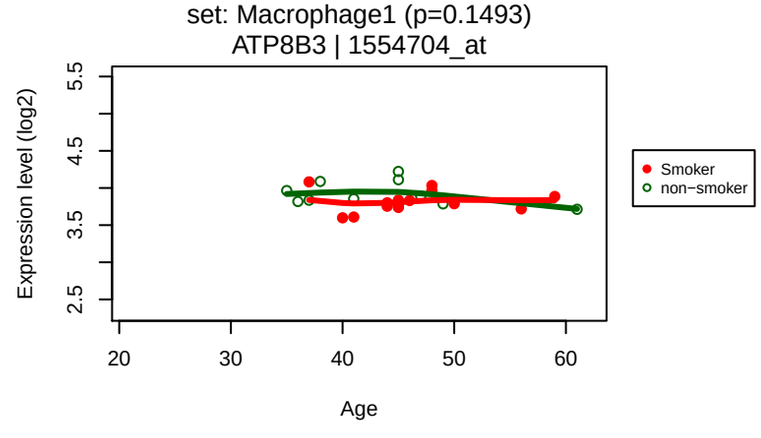

set: SmallAirway1 (p=0.9953)
ATP8B3 | 1554704_at
Expression level (log2)
Age
Smoker
non−smoker

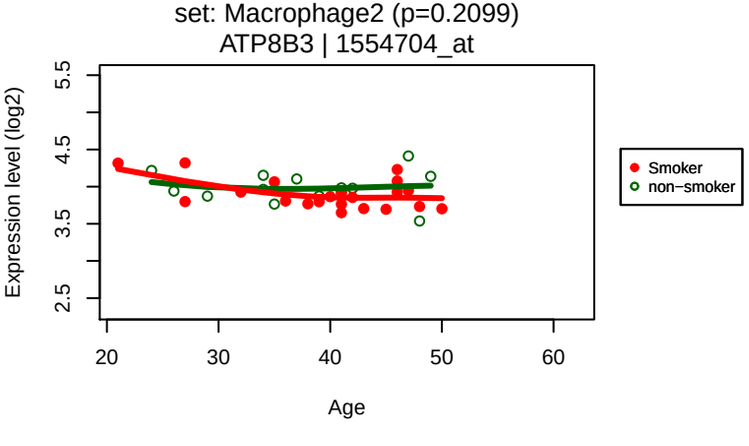

set: SmallAirway2 (p=0.4064)
ATP8B3 | 1554704_at
Expression level (log2)
Age
Smoker
non−smoker

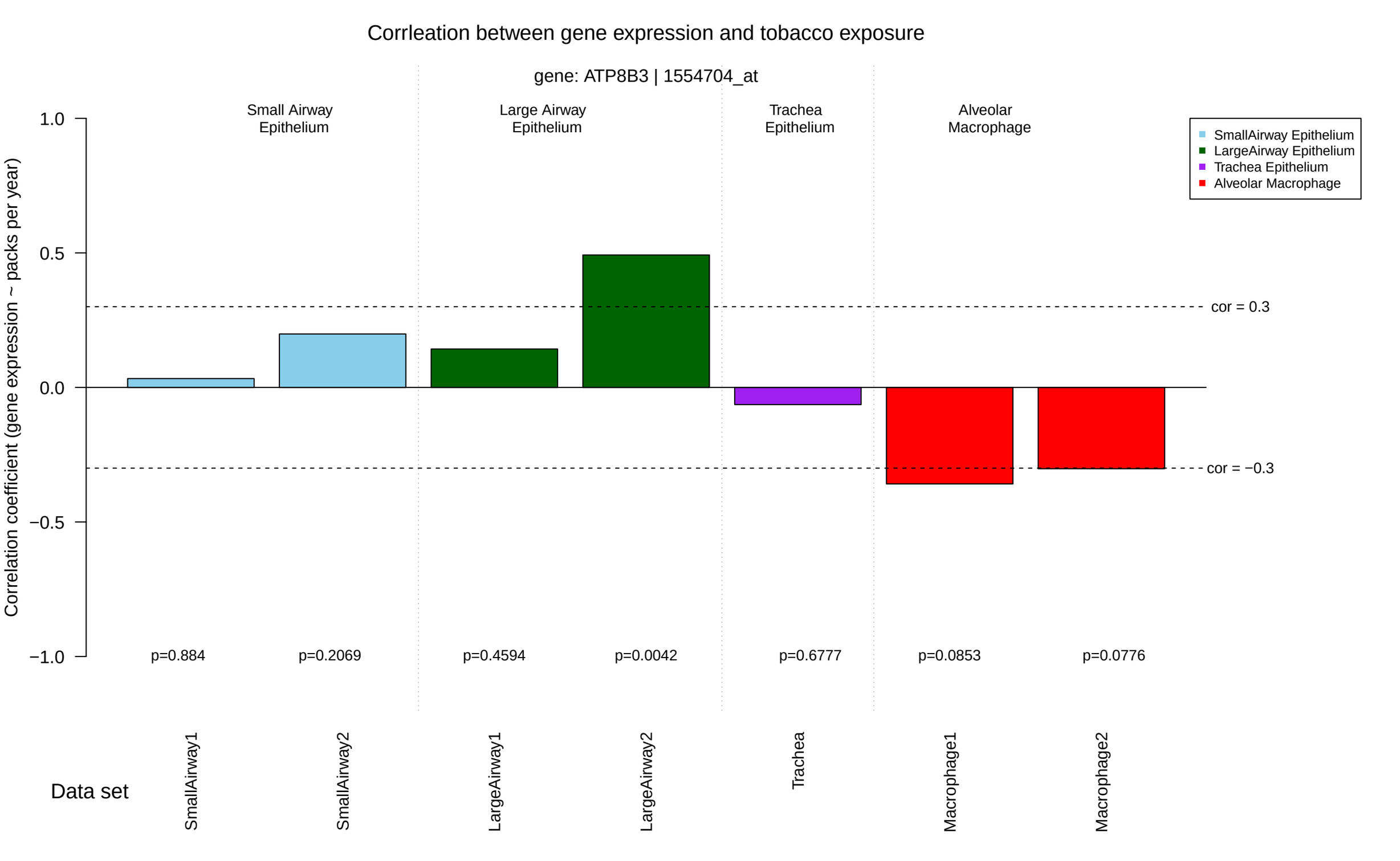

set: LargeAirway1 (p=0.0755)
ATP8B3 | 1554704_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0126)
ATP8B3 | 1554704_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9592)
ATP8B3 | 1554704_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1493)
ATP8B3 | 1554704_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2099)
ATP8B3 | 1554704_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP8B3 | 1554704_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.884
p=0.2069
p=0.4594
p=0.0042
p=0.6777
p=0.0853
p=0.0776
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2